another version of the spinning bucket experiment carried out in deep, cold, completely empty space. We can't run exactly the same experiment, since the shape of the water's surface depended in part on the pull of earth's gravity, and in this version the earth is absent. So, to create a more workable example, let's imagine we have a huge bucket—one as large as any amusement park ride—that is floating in the darkness of empty space, and imagine that a fearless astronaut, Homer, is strapped to the bucket's interior wall. (Newton didn't actually use this example; he suggested using two rocks tied together by a rope, but the point is the same.) The telltale sign that the bucket is spinning, the analog of the water being pushed outward yielding a concave surface, is that Homer will *feel* pressed against the inside of the bucket, his facial skin pulling taut, his stomach slightly compressing, and his hair (both strands) straining back toward the bucket wall. Here is the question: in *totally* empty space—no sun, no earth, no air, no doughnuts, no anything—what could possibly serve as the "something" with respect to which the bucket is spinning? At first, since we are imagining space is completely empty except for the bucket and its contents, it looks as if there simply isn't anything else to serve as the something. Newton disagreed.

He answered by fixing on the ultimate container as the relevant frame of reference: *space itself.* He proposed that the transparent, empty arena in which we are all immersed and within which all motion takes place exists as a real, physical entity, which he called absolute space.[2] We can't grab or clutch absolute space, we can't taste or smell or hear absolute space, but nevertheless Newton declared that absolute space is a something. It's the something, he proposed, that provides the truest reference for describing motion. An object is truly at rest when it is at rest with respect to absolute space. An object is truly moving when it is moving with respect to absolute space. And, most important, Newton concluded, an object is truly accelerating when it is accelerating with respect to absolute space.

Newton used this proposal to explain the terrestrial bucket experiment in the following way. At the beginning of the experiment, the bucket is spinning with respect to absolute space, but the water is stationary with respect to absolute space. That's why the water's surface is flat. As the water catches up with the bucket, it is now spinning with respect to absolute space, and that's why its surface becomes concave. As the